



Blinkit Delivery Delay Root Cause Analysis

Analysis, Diagnosis & Recommendation

1. Executive Summary

The analysis shows that Blinkit's delivery problem is **not isolated to one zone or one operational function**.

Average delivery time is approximately **15 minutes**, while **96.75% of orders exceed the 10-minute SLA**. The problem becomes significantly worse between **20:00 and 23:00**, when average delivery time reaches approximately **20 minutes**.

During these peak hours, several zone-hour combinations show high active-order volumes relative to available riders and pickers. This indicates a **supply-demand capacity imbalance**, although the specific bottleneck varies by zone.

The data therefore points toward **dynamic peak-hour capacity management** rather than a blanket increase in rider headcount.

2. How Severe Is the Problem?

I first looked at overall delivery performance to understand the scale of the issue.

Metric	Result
Total orders	8,000
Delivered orders	7,869
Cancelled orders	131
Average delivery time	~15 min
Cancellation rate	1.64%
10-minute SLA breach	96.75%

Finding

The problem is clearly operationally significant.

Almost **97% of orders are missing the 10-minute SLA**, meaning the issue is not being driven by a small number of extreme delays.

The immediate concern is therefore **delivery reliability**, rather than isolated late orders.

3. Is This a Zone-Specific Problem?

I compared delivery performance across all four zones.

Zone	Avg Delivery Time	SLA Breach
North	14.82 min	96.27%
East	15.00 min	97.03%
South	15.04 min	96.61%
West	15.11 min	97.09%

Finding

The differences between zones are relatively small.

West has the highest average delivery time, while North performs slightly better, but **all four zones have severe SLA breach rates**.

What this tells me

This is a system-wide operational problem rather than a failure isolated to one zone.

This also means that simply fixing one zone is unlikely to resolve the overall delivery problem.

4. When Does the Problem Become Severe?

The biggest clue came from breaking delivery performance down by hour. During most periods, average delivery time remains around the mid-teens. However, between **20:00 and 23:00**, performance deteriorates sharply:

Hour	Avg Delivery Time	SLA Breach
20:00	20.16 min	100%
21:00	19.91 min	100%
22:00	19.96 min	100%
23:00	20.28 min	100%

Finding

The delivery crisis is heavily concentrated in the 20:00–23:00 peak window.

This was an important shift in the investigation.

Instead of asking:

"Which zone is failing?"

the more useful question became:

"What changes operationally during these peak hours?"

5. Store vs Road — Where Is the Bottleneck?

I then compared active orders with available riders and pickers to understand whether the pressure was occurring inside the store or on the road.

South — 22:00

- 115 active orders
- 5 riders
- **23 orders per rider**

West — 23:00

- 149 active orders
- 8 riders
- **18.63 orders per rider**

East — 22:00

- 140 active orders
- 8 riders
- **17.50 orders per rider**

These numbers indicate significant rider capacity pressure during high-load periods.

However, North showed a different constraint.

North — 23:00

- 83 active orders
- 3 pickers
- **27.67 orders per picker**

This suggests that orders can also experience significant pressure **inside the store before dispatch**.

What Does This Tell Us?

The bottleneck is **not identical across every zone**.

Some zone-hour combinations show stronger rider capacity pressure, while others show significant picker pressure.

For example:

- **South → rider pressure**
- **West → rider pressure**

- East → rider pressure
- North → picker pressure

Diagnosis

✦✦ The data suggests a broader supply-demand imbalance, with the specific operational bottleneck varying by zone and hour.

Blinkit needs to match operational capacity with demand at the zone-hour level.

6. Testing the Rain Hypothesis

The Engineering team suggested that seasonal rain may explain the increase in delivery time.

I treated this as a hypothesis rather than assuming it was the cause.

However, the available operational dataset does **not contain weather/rain data**.

Therefore, I cannot directly compare:

┆ Rain periods vs. non-rain periods

or quantify the effect of rain on delivery time.

Conclusion

✦✦ The Rain hypothesis cannot be validated using the available data. I therefore do not attribute the delivery degradation to weather without supporting evidence.

The strongest evidence available in the current dataset points toward **late-evening capacity pressure**, but weather would need to be tested separately if the relevant data becomes available.

7. Is the Customer Funnel Breaking?

I also checked whether the operational problem was creating a major funnel drop-off.

The event progression was:

Funnel Stage	Events
Home	8,000
Search	7,978
PDP	7,957
Cart	7,931
Checkout	7,912
Payment	7,886
Delivered	7,869

The overall progression from Home to Delivered is:

$$7,869 / 8,000 = 98.36\%$$

Finding

There is no single major funnel drop-off.

The largest individual drops are only **26 events** between PDP → Cart and Checkout → Payment.

Business implication

✦✦ **The primary problem is not customer acquisition, browsing, checkout, or payment conversion. The major issue occurs in the operational delivery experience after the order is placed.**

This means I would not prioritize funnel optimization as the immediate response to this problem.

8. Customer Cancellations

The order data shows:

Status	Orders	Share
Delivered	7,869	98.36%

Status	Orders	Share
Cancelled	131	1.64%

Finding

The current cancellation rate is **1.64%**.

This is much smaller than the 96.75% SLA breach rate, which suggests that the delivery problem has **not yet translated into widespread immediate cancellation** in the available dataset.

However, continued failure to meet the promised delivery time can still create a longer-term risk to:

- Customer trust
- Retention
- Repeat purchases
- Perceived reliability

Important limitation

I cannot reliably identify the exact funnel stage where cancellations occur from the current event data because `blinkit_orders` does not contain a `session_id` to connect each order to its specific customer session.

Therefore, I would not claim that cancellation is concentrated at checkout, payment, or any other specific stage.

9. Final Diagnosis

The data suggests the root cause is...

■ **A late-evening supply-demand capacity imbalance.**

The problem affects all four zones, but becomes particularly severe between **20:00 and 23:00**, when delivery times increase to approximately 20 minutes.

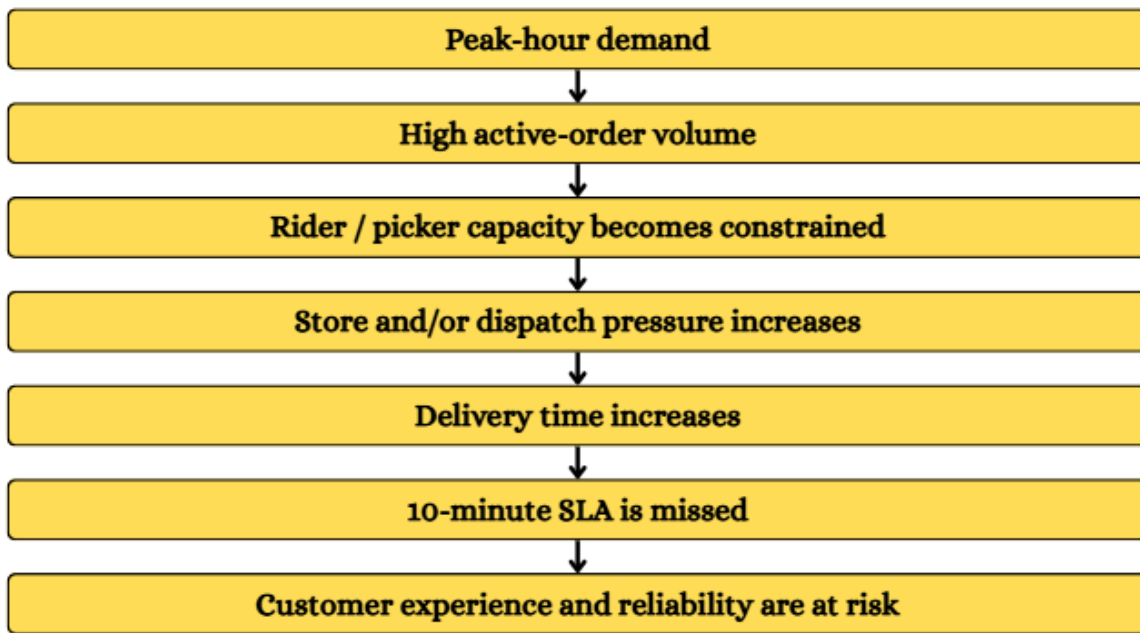
During these periods, active orders become high relative to available rider and picker capacity.

The important finding is that the bottleneck is **not uniform**:

- Some zones experience stronger rider pressure.

- Other zones experience stronger picker pressure.
- All zones are affected at the network level.

Root Cause Chain



10. Recommendation

- ✦ ✦ **Implement zone-hour capacity planning for the 20:00–23:00 peak period, dynamically reallocating riders toward zones with high orders-per-rider and increasing picker coverage where orders-per-picker indicates store congestion.**

Capacity should be allocated based on **where and when demand is highest**.

Example

If the system detects:

South — high orders/rider

→ Reallocate additional riders.

North — high orders/picker

→ Increase picker coverage.

This creates a more targeted response than simply adding a fixed number of riders across the network.

11. Why This Recommendation?

The analysis suggests that the problem is driven by **capacity mismatch**, not simply a shortage of one resource.

A blanket hiring approach could increase costs without solving the actual bottleneck in every zone.

A dynamic approach allows Blinkit to:

- Respond to demand spikes
- Target the most constrained zones
- Improve peak-hour utilization
- Reduce unnecessary capacity costs
- Intervene before SLA performance deteriorates

12. Risks & Trade-offs

The main risks to weigh are:

Higher operating cost

Additional peak-hour riders or pickers will increase labor and operational costs.

Reallocation risk

Moving riders between zones could create shortages elsewhere if demand forecasts are inaccurate.

Under-utilization

If demand falls unexpectedly, additional capacity may remain underutilized.

Operational complexity

Dynamic capacity allocation requires reliable real-time monitoring and forecasting.

The solution should therefore be **data-driven and monitored continuously**, rather than implemented as a permanent blanket increase in capacity.

13. Success Metrics

I would measure the success of the intervention using three primary metrics.

1. 10-Minute SLA Adherence Rate

Goal: Increase significantly.

This is the primary measure of whether Blinkit is restoring its core delivery promise.

2. Average Delivery Time

Goal: Move closer to the 10-minute SLA.

3. 20:00–23:00 SLA Breach Rate

Goal: Reduce significantly during the peak window.

I would also monitor **cancellation rate** as a secondary customer-health metric.

14. Final Business Recommendation

✦✦ **Blinkit should prioritize dynamic zone-hour capacity management rather than applying a blanket increase in rider headcount. The strongest operational signal is the sharp deterioration between 20:00 and 23:00, when high order volumes place pressure on rider and picker capacity. By reallocating capacity toward the specific zone-hour combinations experiencing the greatest pressure, Operations can target the bottleneck directly while controlling unnecessary operating costs.**

15. Executive Takeaway

✦✦ **The delivery crisis is not a single-zone problem and there is no single universal bottleneck. The data points to a late-evening capacity**

imbalance affecting both rider and store operations, with the exact constraint varying by zone and hour. The immediate priority should therefore be to dynamically match rider and picker capacity with demand during the 20:00–23:00 peak, while monitoring SLA adherence and delivery time to measure the impact.